Worksheet 5: Cubes, Primes, and Number Properties

Topic Focus: Advanced number families, cube numbers, prime numbers, and patterns in operations.

1. The numbers 1, 8, 27, 64, 125 are known as:
A) Square numbers B) Cube numbers C) Triangular numbers D) Virahanka numbers
Hint: Created by multiplying a number by itself three times (n x n x n).

2. Which of these is the cube of 6 (6x6x6)?
A) 36 B) 128 C) 216 D) 256
Hint: 6 x 6 = 36. 36 x 6=216.

3. Every prime number except for one is an odd number. Which prime number is even?
A) 1 B) 2 C) 4 D) No even primes
ans: 2 only has factors of 1 and itself. And 2 is also even.

4. Pattern of adding odd numbers: 1=1; 1+3=4; 1+3+5=9. Sum of 1+3+5+7+9?
A) 16 B) 20 C) 25 D) 36
Hint: Adding the first N odd numbers always gives the Nth square number. 5 odd numbers = 5^2.

5. Which number is missing in this sequence of cubes? 1, 8, __, 64, 125.
A) 16 B) 24 C) 27 D) 36
Hint: This is the missing 3^3. (3 x 3 x 3).

6. Look at the last digits of square numbers (1, 4, 9, 16, 25...). Which digit NEVER ends a perfect square?
A) 1 B) 4 C) 5 D) 8
Hint: Perfect squares can only end in 0, 1, 4, 5, 6, or 9. Never 2, 3, 7, or 8.

7. A Magic Square has rows, columns, and diagonals that add up to the same number. In a 3x3 square using 1-9, what is the magic sum?
A) 12 B) 15 C) 18 D) 21
Hint/solu: The sum of 1 through 9 is 45. Divided equally among 3 rows, 45/3 = 15.

8. Sequence: 1000, 100, 10, 1. What comes next?
A) 0 B) 0.1 C) -10 D) -1
hint: Dividing by 10.

9. The sum of two odd numbers is always:
A) Odd B) Prime C) Even D) Square
Hint: (2n+1) + (2m+1) = 2(n+m+1). Adding two leftover 1s makes an even pair.

10. Multiply any number by 9, add digits until a single digit remains (eg, 9x12=108, 1+0+8=9). Pattern?
A) Sum is always 9 B) Sum alternates 3 and 9 C) Sum increases by 1 D) Sum is even
Hint: This is a unique property of the number 9 (digital root of multiples of 9 is 9).

11. Prime numbers that have a difference of 2 (like 3 and 5, or 11 and 13) are called:
A) Double primes B) Twin primes C) Co-primes D) Even primes
Concept: They are twins because they are as close together as two odd primes can be.

12. If a number ends in 0, its cube (n x n x n) must end in how many zeros?
A) 1 B) 2 C) 3 D) 4
ans: 10 x 10 x 10 = 1000. It triples the amount of zeros.

13. Which of the following is a Perfect Number (a number equal to the sum of its proper divisors)?

A) 6 B) 10 C) 12 D) 15

ans: Factors of 6 (excluding 6) are 1, 2, and 3. 1 + 2 + 3 = 6.

14. Two numbers that have NO common factors other than 1 (like 8 and 15) are called:

A) Twin primes B) Co-primes C) Composite pairs D) Magic numbers

hint: Neither 8 nor 15 are prime themselves, but co-operatively they share no factors.

ans: composite no.

15. What is the pattern in the differences between consecutive cubes (8-1=7, 27-8=19, 64-27=37)?

A) Multiples of 6 plus 1 B) Prime numbers C) Even numbers D) Square numbers

pattern: 7 = 6(1)+1. 19 = 6(3)+1. 37 = 6(6)+1. (formula: Difference is 3n^2 + 3n + 1).